

Design and Optimization of a Conformal 433 MHz Helmet-Mounted Antenna for Tactical Application

Noor Amalina Ramli

Department of Electrical and Electronic Engineering
Universiti Pertahanan Nasional Malaysia
Kuala Lumpur, Malaysia
nooramalina@upnm.edu.my

Mohd Saiful Riza Bashri

Department of Sciences, Kulliyah of Engineering
International Islamic University Malaysia,
Kuala Lumpur, Malaysia
mohdsaifulriza@iiium.edu.my

Abstract—This study investigates the design and simulation of a conformal antenna operating at 433 MHz for integration onto a military helmet. The antenna is optimized to follow the curved surface of the helmet, enabling low-profile deployment for soldier-worn communication systems. Full-wave electromagnetic simulations were conducted using CST Studio Suite to evaluate key performance parameters, including the reflection coefficient (S_{11}), radiation pattern, and antenna gain. The results demonstrate good impedance matching at the target frequency, with a stable radiation pattern and acceptable gain despite the curvature and proximity to the human head. These findings indicate that the proposed helmet-mounted conformal antenna is a promising candidate for reliable tactical communication at 433 MHz, with potential for further improvement and real-world deployment.

Keywords—conformal antenna, monopole antenna, specific absorption rate (SAR).

I. INTRODUCTION

Reliable and efficient communication is a critical requirement in modern military operations, where real-time coordination, situational awareness, and data exchange are essential for mission success. To support these needs, wearable communication systems that can be seamlessly integrated into a soldier's gear are becoming increasingly important. One promising approach is to mount antennas directly onto military helmets, enabling hands-free operation while minimizing the impact on mobility and comfort.

The 433 MHz frequency, part of the Industrial, Scientific, and Medical (ISM) band, is widely used for short-range, low-power wireless communication. Its relatively low frequency allows for better propagation through obstacles and around the human body, making it suitable for tactical environments. However, designing an antenna at this frequency that is both compact and conformal to the curved helmet surface presents significant challenges. These include detuning caused by the helmet material and the proximity of the human head, reduced radiation efficiency due to body absorption, and pattern distortion from the non-planar mounting surface.

Conformal antennas, which follow the shape of the mounting surface, offer a practical solution to these challenges. They provide a low-profile form factor that reduces visual signature and physical interference while maintaining effective radiation characteristics. Advances in electromagnetic simulation tools have made it possible to design and optimize such antennas without the immediate need for physical prototypes.

Several recent studies have explored key aspects of wearable antenna design. Bashri et al. [1] analysed the impact

of head curvature and antenna proximity on performance parameters. Gopi et al. [2] investigated the size constraints and efficiency limitations of 433 MHz body-worn antennas. Ibrahim et al. [3] proposed meander-line structures for space-constrained wearable systems. Missaoui and Kaddour [4] demonstrated that flexible substrates such as liquid crystal polymer can sustain performance under mechanical deformation. Tang et al. [5] introduced a helmet-based hemispherical-conformal array with dual circular polarizations and wide-angle beam coverage, highlighting the feasibility of advanced wearable antennas for wireless body area networks in emergency and military scenarios. Additionally, research on specific absorption rate (SAR) highlights the importance of minimizing electromagnetic exposure to the head, reinforcing the need for SAR-aware design in wearable systems.

Despite these contributions, there remains limited work that specifically addresses the design and optimization of a low-profile, 433 MHz conformal patch antenna intended for integration onto military helmets, particularly with emphasis on both electromagnetic performance and SAR compliance. This gap presents a barrier to the practical deployment of low-frequency, helmet-mounted communication systems in tactical environments.

This paper aims to address this gap by presenting the design, simulation, and SAR evaluation of a 433 MHz conformal patch antenna mounted on a ballistic helmet. The antenna is first designed in a planar configuration and subsequently conformed to a realistic helmet model. Key performance metrics including reflection coefficient, gain, radiation pattern, and SAR distribution are evaluated through full-wave simulation in both free-space and on a human head phantom.

The objective is to assess the feasibility of integrating such an antenna into soldier-worn systems and to provide a validated simulation framework that can support future prototyping and field deployment. Unlike existing 433 MHz wearable antenna studies, this workplace specific emphasis on conformal integration with a ballistic helmet, taking into account realistic conditions and human body interactions. It also includes SAR evaluation across different power levels, which is often overlooked in prior research. These additions make the proposed design more relevant and applicable for real-world tactical communication systems.

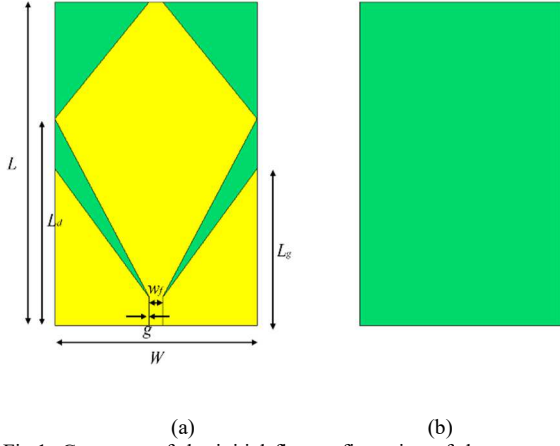


Fig.1. Geometry of the initial flat configuration of the proposed monopole antenna. (a) Top view and (b) bottom view.

II. ANTENNA DESIGN

This section describes the theoretical foundation and simulation setup used to design a compact patch antenna operating at 433 MHz, intended for conformal mounting on a ballistic helmet. All designs and performance evaluations were carried out using Microwave CST Studio Suite, a full-wave electromagnetic solver based on the finite integration technique (FIT).

The diamond shape was chosen over conventional rectangular monopole or patch geometry due to its longer current path for a given footprint, which is beneficial at low frequencies like 433 MHz. This shape also provides better impedance matching flexibility and help maintain compactness which is an important factor when conforming the antenna to a limited surface area on the helmet.

A. Flat Configuration

In the initial stage of the design process, a planar monopole with diamond shape operating at 433 MHz was developed as shown in Fig. 1. The antenna was initially simulated on ultra-high molecular weight polyethylene (UHMW PE) substrate which is the common material used in military helmet [6]. Key design parameters such as patch dimensions, ground plane size, and feed location were optimized to achieve a good impedance match and acceptable radiation characteristics in free space.

The length of the monopole antenna, L was calculated based on (1)

$$L = 0.25 \times \lambda_g \quad (1)$$

where the guided wavelength, $\lambda_g = 1/\sqrt{\epsilon_r}$. The dielectric constant of the substrate, ϵ_r is 2.3 [6]. The feeding line is made of co-planar waveguide (CPW) structure where its characteristic impedance is 50Ω . The thickness of the initial flat antenna structure is set to 2mm. To account for fringing effects and to improve accuracy, the design was refined using CST's parametric sweep feature. The simulation environment was set up with open boundary conditions to mimic free-space operation, and key antenna characteristics such as

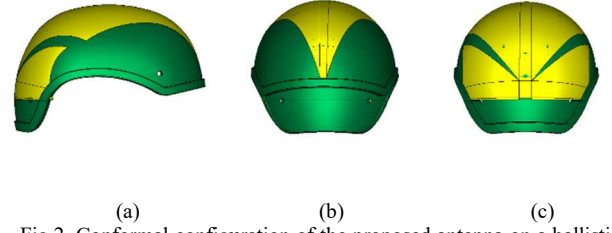


Fig.2. Conformal configuration of the proposed antenna on a ballistic helmet. (a) Side view, (b) front view and (c) back view.

reflection coefficient, gain, and far-field radiation pattern were analyzed. These results served as a baseline to evaluate how the antenna would perform under more realistic mounting scenarios involving curvature and proximity to the human body. The flat antenna parameters are given in Table I.

TABLE I. GEOMETRIC PARAMETER OF THE FLAT ANTENNA

Antenna Parameters	Symbol	Value(mm)
Antenna length	L	227
Antenna width	W	150
Co-planar waveguide gap	g	0.2
Ground length	L_g	110
Patch center length	L_d	145
Feeding line width	w_f	10

B. Conformal Configuration

After establishing the performance of the flat monopole antenna, the same structure was adapted to a conformal configuration by mounting it onto the curved surface of a ballistic helmet. The helmet model was imported into CST as a solid geometry, and the antenna was reshaped to follow the surface curvature using CST's built-in transformation tools as shown in Fig. 2.

The presence of curvature affects several aspects of antenna performance. The change in geometry can shift the resonant frequency, introduce impedance mismatches, and modify the current distribution across the patch. To account for these effects, the antenna length, L was re-optimized in the conformal layout. The helmet material was modelled with a realistic dielectric constant, representing materials such as ABS or Kevlar commonly used in ballistic helmets. Special attention was given to mesh generation and resolution to ensure accurate modelling of the complex surface and field interactions.

Simulations were then performed for the conformal antenna in the same free-space environment, using identical excitation and boundary settings as in the flat case. Reflection coefficient, gain, and radiation patterns were extracted and compared with the flat configuration to assess the impact of mounting curvature. This evaluation helped identify how much performance was altered due to the realistic placement on the helmet and provided guidance on whether further design adjustments were needed to compensate for detuning or pattern distortion.

III. SIMULATION WITH HEAD PHANTOM

To evaluate the antenna's performance in a more realistic use case, the conformal monopole antenna was simulated in the presence of a human head phantom as illustrated in Fig. 3. The objective of this simulation was to assess how proximity to human tissue affects the antenna's electromagnetic performance, particularly in terms of impedance matching, radiation efficiency, and field distribution. The presence of biological tissue near the antenna can lead to detuning, increased power absorption, and alterations in the far-field radiation pattern, which are critical considerations for wearable and body-mounted communication systems.

A human head model was used in the simulation, selected from CST's built-in human body models. This model includes layered representations of different biological tissues, such as skin, bone, and brain, each assigned frequency-dependent dielectric properties derived from published data. The electrical parameters, including relative permittivity and conductivity, were automatically applied from CST's biological materials library for 433 MHz.

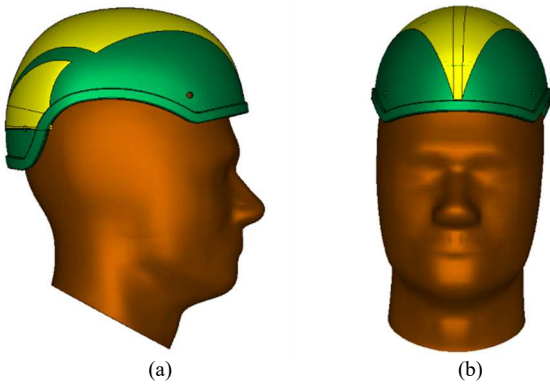


Fig.3. Helmet-mounted antenna with head phantom. (a) Side view and (b) front view.

The conformal antenna was positioned on the surface of the ballistic helmet, which was then aligned appropriately on the head phantom. Realistic distances and orientation between the antenna, helmet, and phantom were set in the simulation. The simulation domain remained the same as in the previous steps, with open boundary conditions and the same excitation port configuration to allow for direct comparison.

The simulation outputs focused on key performance indicators, including reflection coefficient, gain, and three-dimensional radiation pattern. Additional emphasis was placed on evaluating near-field distributions, especially around regions of the head closest to the antenna. Any observable frequency shifts, reductions in gain, or significant pattern changes were noted and compared to the results from the free-space, helmet-only configurations and with head phantom. These observations provide insight into the effects of human-body loading on the antenna and help determine whether the design remains suitable for practical deployment near the user.

IV. SIMULATION OF SAR WITH VARYING TRANSMITTED POWER LEVEL

In addition to evaluating electromagnetic performance, it is essential to assess the safety of wearable antennas operating near the human body. One of the key metrics used for this purpose is the specific absorption rate (SAR), which quantifies the rate at which electromagnetic energy is absorbed by biological tissues. SAR analysis ensures compliance with international safety guidelines and is especially important for body-worn systems where prolonged exposure may occur.

SAR simulations were carried out in CST Studio Suite using the same head phantom and helmet-mounted antenna configuration described in the previous section. The simulation setup followed the IEEE C95.1 and ICNIRP standards, which define acceptable SAR limits for both localized and whole-body exposure. CST's SAR calculation module was used to evaluate the 1-gram averaged SAR values across the tissue regions closest to the antenna.

To understand the relationship between transmitted power and SAR levels, the simulations were repeated using several power levels: 100mW, 300mW, and 500 mW. This range was selected to represent low- to medium-power wireless systems commonly used in tactical short-range communication. The antenna excitation was scaled accordingly, and all other simulation parameters were kept consistent to ensure a controlled comparison.

The highest SAR values were typically observed in regions near the ear and temple, where the antenna was positioned closest to the tissue. The simulation results showed that at power levels up to 500mW, 1g SAR values remained well below the recommended limits of 1.6 W/kg. These results provide critical insight into the safe operating range of the proposed antenna. While the design meets safety standards at moderate power levels, further mitigation measures such as substrate selection, shielding, or power control may be required for higher-power operation. The SAR analysis thus complements the electromagnetic performance evaluation by identifying operating conditions that balance both performance and user safety.

V. RESULT AND DISCUSSIONS

This section presents the results of the simulations and discusses the observed effects of conformal integration, proximity to the human head, and varying transmit power levels on antenna performance and user safety.

In the flat configuration, the patch antenna achieved good impedance matching with an S_{11} value of -14.837 dB at the target frequency of 433 MHz as shown in Fig. 4. The far-field radiation pattern showed a broad main lobe with moderate gain of approximately 2.12dBi indicating suitability for omnidirectional short-range communication. These results confirmed the design validity in free space and served as a baseline for later comparison.

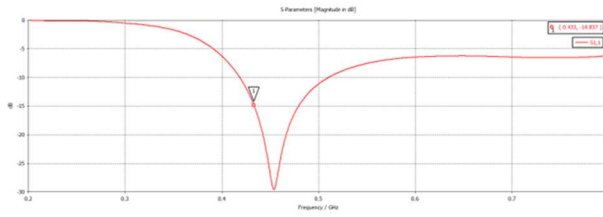


Fig.4. Simulated reflection coefficient of the flat antenna configuration.

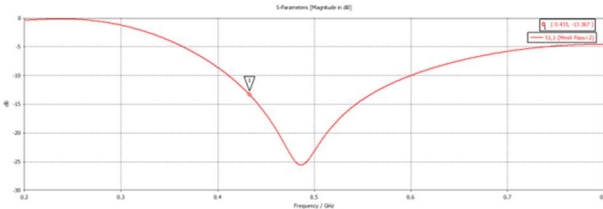


Fig.5. Simulated reflection coefficient of the conformal antenna configuration (helmet only case).

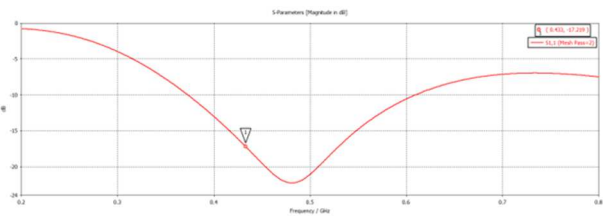


Fig.6. Simulated reflection coefficient of helmet-mounted antenna with head phantom.

When conformed to the helmet surface, the antenna maintained acceptable performance but experienced detuning, with the resonant frequency shifting upward 791.23MHz. This was attributed to the change in effective dielectric loading and current path due to surface curvature. The length of the antenna was increased from 227mm to 345mm to shift the operating frequency back to 433MHz. The reflection coefficient after the re-optimization of the antenna length is shown in Fig. 5. The radiation pattern exhibited minor asymmetry but remained functional. The gain decreased slightly to 2.04dBi. Simulation with the human head phantom revealed further performance changes. However, only slight detuning was observed compared to helmet-only case. The impedance matching remained within acceptable limits as confirmed by the reflection coefficient result shown in Fig. 6. The peak gain somehow increased to 3.84dBi and three-dimension radiation pattern shows similar performance to helmet only case. The three-dimension

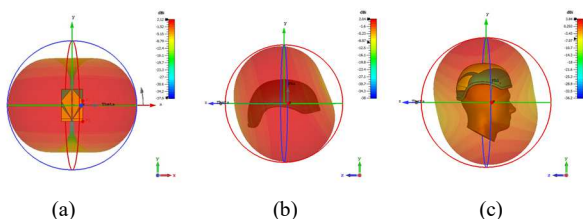


Fig.7. 3-D radiation pattern of the proposed antenna. (a) Flat antenna, (b) conformal antenna (helmet only case) and (c) helmet-mounted antenna with head phantom.

radiation patterns for all three cases are illustrated in Fig. 7. These results highlight the importance of including human-body interaction in the design and evaluation process for wearable antennas. SAR simulations showed that power absorption was concentrated in tissue regions directly beneath or adjacent to the antenna. At a transmitted power of 100 mW, the maximum 1g SAR was 0.152W/kg, which remained within the safety threshold defined by IEEE C95.1. As the transmit power increased to 300mW and 500mW, the SAR values increase to 0.457W/kg and 0.762W/kg respectively. These findings suggest that the antenna is safe for use up to 500mW of transmitted power levels. Integrating shielding layers may further reduce SAR without compromising antenna performance.

VI. CONCLUSIONS

This paper presents the design, simulation, and safety evaluation of a conformal 433 MHz microstrip patch antenna intended for integration onto a ballistic military helmet. Using Microwave CST Studio Suite, a flat antenna was first designed and optimized in free space to serve as a performance baseline. The antenna was then conformed to a realistic helmet surface, followed by simulations involving a human head phantom to capture the effects of proximity to biological tissue. The results show that the antenna maintains acceptable performance in both conformal and realistic use conditions, with only moderate reductions in gain and minor detuning observed. Simulations with the head phantom confirmed that while body proximity impacts efficiency and radiation pattern, the antenna continues to operate effectively at desired frequency. Future work involves physical prototyping and experimental validation, as well as potential integration into soldier-worn communication systems.

REFERENCE

- [1] Mohd. S. R. Bashri, T. Arslan, and W. Zhou, "Flexible Antenna Array for Wearable Head Imaging System," IEEE, 2017.
- [2] D. Gopi, A. R. G. Chandra, K. Mouli, V. Janardhan Reddy, G. K. Chaitanya, and S. Ramu, "Miniaturized cpw-fed conformal antenna for guided missiles," in *Journal of Physics: Conference Series*, IOP Publishing Ltd, Dec. 2020. doi: 10.1088/1742-6596/1706/1/012073.
- [3] H. H. Ibrahim *et al.*, "Low Profile Monopole Meander Line Antenna for WLAN Applications," *Sensors*, vol. 22, no. 16, Aug. 2022, doi: 10.3390/s22166180.
- [4] S. Missaoui, "Liquid crystal-reconfigurable conformal monopole antennas structures for microwave applications," *Journal of Engineering and Technology Research*, vol. 4, no. 3, Feb. 2012, doi: 10.5897/jetr11.080.
- [5] W. Tang *et al.*, "A Helmet-Based Hemispherical-Conformal Wearable Array Antenna with Broad Bandwidth, Wide-Angle Coverage and Reconfigurable Dual Circular Polarizations for WBAN Applications," *IEEE Trans Antennas Propag*, 2025, doi: 10.1109/TAP.2025.3560181.
- [6] "Ultra-High Molecular Weight Polyethylene & Combat Helmets – Hard Head Veterans." Accessed: Jun. 30, 2025. [Online]. Available: <https://www.hardheadveterans.com/blogs/reviews/uhmw-pe-and-the-modern-helmet>